3. Individual Dental & Vision Cashflow & Pent-Up Demand Engine — Humana NPV Model (v4.2.0)

Actuarial Overview (Short-Duration Cohort, 10-Year Horizon)

In individual Dental & Vision (modeled per 1,000 individual policies over 10 years / 120 monthly rows), the pricing and forecasting actuary faces **"buy, use, and lapse" anti-selection**:

- **Year-1 Pent-Up Demand:** Newly enrolled members catch up on delayed dental procedures in their first 12 months of eligibility (`DV_PentUp = 1.28×`).
- **High Attrition:** Annual lapse rates run at roughly 30% (`DV_Lapse1..3`), meaning policies exit before Humana can recover upfront acquisition costs and Year-1 claim strain.
- **Unprofitable on Base Market Inputs:** Using market premiums ($468/yr from EMI filings), market loss ratios (Milliman NADP survey), and 30% lapse from filed memoranda, the base product loses money (`Cash ΠPV = -$50,128`, `DE ΠPV = -$40,686`, `P(DE < 0) = 99.8%`).

Visual Cashflow & Benefit Category Architecture

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                        DENTAL & VISION: 10-YEAR (120-MONTH) CASHFLOW ENGINE
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 [ISSUE: t = 0] ──▶ Outflow: −1,000 × DV_Acq
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    ▼
  [PERSISTENCY ENGINE (Each Month k = 0 ... 119)]
  • No separate mortality decrement modeled (short-duration product; bundled into
total termination).
  • High early lapse: q_annual = min(0.95, Lapse_by_Duration(Yr 1, 2, 3+) ×
DV_LapseMult)
  • Exposure = InForce_Start (since lapses occur at month-end after premium is paid in
advance).

  [BENEFIT CATEGORY & DEFERRED PENT-UP DEMAND ENGINE (4 Categories: Preventive, Basic,
Major, Vision)]
  For each category c = 1..4 with Cost Share S_c, Trend Add-on TA_c, Waiting Period
W_c (months),
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  and normalized Pent-Up Relativity PR_c:
  • If month k < W_c (During Waiting Period): Category claim cost = $0.
  • If W_c ≤ k < W_c + 12 (First 12 Months of Eligibility):
    Pent-up demand is DEFERRED, not eliminated! Multiplier = DV_PentUp × PR_c.
  • If k ≥ W_c + 12 (After First 12 Months of Eligibility): Multiplier = 1.0.

  [MONTHLY CASHFLOWS]
  (+) Premium (Start of Month):
      InForce_Start × [DV_Prem1 × (1 + DV_RateInc)^(Yr-1)] ÷ 12
  (-) Commissions & Tax (Start of Month):
      Premium × [IF(Yr=1, DV_FYComm, DV_RenComm) + DV_PremTax]
  (-) Incurred Claims (Mid-Month):
      Exposure × DV_Clm1 × SUM over c=1..4 [ S_c × (1 + DV_Trend + TA_c)^(Yr-1) ×
PentUp_c(k) × Elig_c(k) ] ÷ 12
  (-) Maintenance Admin (Mid-Month):
      Exposure × [DV_Maint × (1 + ExpInfl)^(Yr-1)] ÷ 12
  (•) DAC Tax: 0% in base case (treated as excepted-benefit cancellable; swept at 9.2%
if GR).
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Step-by-Step Mathematical Equations (DentalVision_Monthly Columns A to T)

Let month index k run from 0 to 119:

- **Policy Year (Col B):** Year(k) = INT(k ÷ 12) + 1
- **Month of Policy Year (Col C):** MonthOfYr(k) = (k mod 12) + 1

1. Monthly Persistency & Exposure Roll-Forward (Cols D–H)

- **Annual Lapse Rate (Col D):**
  q_lapse_annual(k) = min(0.95, DV_Lapse(min(Year(k), 3)) × DV_LapseMult)
- **Monthly Compound Lapse Rate (Col E):**
  q_lapse_monthly(k) = 1 – (1 – q_lapse_annual(k))^(1/12)
- **In-Force at Start of Month (Col F):**
  • Month 0: InForce(0) = DV_Cohort (1,000 policies)
  • Month k: InForce(k) = InForce(k – 1) – Lapses(k – 1)
- **End-of-Month Lapses & Exposure (Cols G–H):**
  Lapses(k) = InForce(k) × q_lapse_monthly(k)
  Exposure(k) = InForce(k) *(Members who lapse at month-end have paid for and are covered during month k.)*

2. Benefit Categories, Waiting Periods & Deferred Pent-Up Demand (Col J)

Version 4.1 models four distinct benefit categories c in {1: Preventive, 2: Basic, 3: Major, 4: Vision}:

- **Category Cost Shares:** S_1, S_2, S_3, S_4 (where S_1 + S_2 + S_3 + S_4 = 100%; Major services S_3 = 24.9%).
- **Normalized Pent-Up Relativities:** Raw category pent-up relativities DV_CatPU_c are normalized so the cost-share-weighted average equals 1.0000:
  DV_CatPR_c = DV_CatPU_c ÷ (S_1 × DV_CatPU_1 + S_2 × DV_CatPU_2 + S_3 × DV_CatPU_3 + S_4 × DV_CatPU_4)
- **Waiting Period & Pent-Up Deferral Rules (v4.1 Fix):**
  In v4.0, a waiting period of W_c months simply zeroed out claims during k < W_c, artificially erasing Year-1 pent-up demand. Version 4.1 defers pent-up demand into the **first 12 months after the waiting period expires** (W_c ≤ k < W_c + 12):
  - **Eligibility Factor Elig_c(k):**
    Elig_c(k) = 1 if k ≥ W_c, else 0 (during waiting period)
  - **Pent-Up Multiplier Pent_c(k):**
    Pent_c(k) = (DV_PentUp × DV_CatPR_c) if (W_c ≤ k < W_c + 12), else 1.0
- **Annualized Claim Cost per Policy in Month k (Col J):**
  ClmRate(k) = DV_Clm1 × SUM over c = 1..4 of [S_c × (1 + DV_Trend + DV_CatTA_c)^(Year(k) − 1) × Pent_c(k) × Elig_c(k)]

3. Monthly Cashflows & Present Value (Cols K–T)

- **Monthly Premium Billed in Advance (Col K):**
  Premium(k) = InForce(k) × [DV_Prem1 × (1 + DV_RateInc)^(Year(k) − 1)] ÷ 12
- **Monthly Incurred Claims (Col L):**
  Claims(k) = Exposure(k) × ClmRate(k) ÷ 12
- **Commissions, Premium Tax, and Maintenance (Cols M–O):**
  • Commission(k) = Premium(k) × (DV_FYComm if Year(k) = 1, else DV_RenComm)
  • PremTax(k) = Premium(k) × DV_PremTax
  • Maint(k) = Exposure(k) × [DV_Maint × (1 + ExpInfl)^(Year(k) − 1)] ÷ 12
- **Monthly Present Value (Col R):**
  PV_Cash(k) = [Premium(k) − Commission(k) − PremTax(k)] × (1 + DiscRate)^(−k ÷ 12) − [Claims(k) + Maint(k)] × (1 + DiscRate)^(−(k + 0.5) ÷ 12)

Valuation Results, Reverse Stress & Product Design Remedies

Scenario / Product Option	Cash NPV (per 1,000)	DE NPV (After Tax & Capital)	DE vs. Base	PV Loss Ratio	Actuarial Interpretation
Base Case (0-Month Wait on All Categories)	**-$50,128**	**-$40,686**	$0	77.6%	Acquisition strain + Year-1 pent-up demand is not recovered before 30% annual lapse.
Reverse Stress: Break-Even Year-1 Premium	+$1,555	**$0 (Break-Even)**	+$40,686	73.8%	Requires a **+5.23% premium increase** ($468/yr → $492.48/yr) to break even on DE NPV. Break-even is not the price: see the v4.2 risk-based recommendation below.
Product Option: 12-Month Wait on MAJOR Services	**+$23,676**	**+$17,493**	**+$58,179**	71.1%	Deferring Major services (24.9% of cost) by 12 months turns the product profitable even after pent-up demand shifts to Year 2 (because 30% of policies lapse

Scenario / Product Option	Cash NPV (per 1,000)	DE NPV (After Tax & Capital)	DE vs. Base	PV Loss Ratio	Actuarial Interpretation
					before Month 12).
Corporate Tax Sensitivity: If Subject to 9.2% DAC Tax	−$50,128	**-$49,502**	-$8,816	77.6%	If individual dental is classified as guaranteed renewable under IRC §848, DAC tax costs an extra -$8,816 of DE.

Critical Actuarial Takeaways for the Interview

1. **Why Unnormalized D&V Trend Was Biased by −1.75 pt:** Because Year-1 policies have a 1.28× pent-up demand load, as an open D&V book matures and the share of duration-1 members declines, raw aggregate PMPM growth appears artificially low. In `02_claims_trend_forecast.py`, failing to normalize for duration/pent-up wear-off biased the D&V trend estimate by −1.75 percentage points.
2. **Upper-Bound Caveat on Waiting Periods:** The +$17,493 DE NPV for a 12-month Major waiting period is an upper bound because lapse rates and buyer anti-selection are held constant; in practice, imposing a waiting period can alter sales mix and Month-13 lapse behavior.
3. **v4.2 Risk-Based Pricing Recommendation:** Break-even (+5.23%) still leaves about a coin-flip chance of missing the hurdle. Priced so that P(DE NPV < 0) ≤ 10%, the premium rises +8.0%, from $39.00 to $42.11 a month ($468.00 → $505.32 a year): P(DE < 0) falls from 99.8% to 9.8% and DE NPV becomes +$21,335 per 1,000 (PV loss ratio 71.9%). Flagged REVIEW because it is 8% above the one public reference rate, so it is really a conversation about acquisition cost and benefit design, not just rate.
4. **Price-Response Robustness:** Lapse does not respond to price in the model. If a higher price raised lapse by 10%, the required premium would be about +9.4% instead of +8.0%.